

1. INTRODUCTION

Automated resume screening is critical for HR teams processing hundreds of applications per role. This project builds a full ML pipeline — from raw CSV to interactive dashboard — to predict resume–job compatibility scores and surface candidate segments through unsupervised clustering.

2. BACKGROUND & LITERATURE REVIEW

- Early IR approaches (TF-IDF, BM25) suffer from vocabulary mismatch.
- Ontology-based skill mapping (Siting et al., 2012) improved recall.
- Ensemble trees (Fernández-Delgado, 2014) excel on tabular HR data.
- XGBoost (Chen & Guestrin, 2016): 2nd-order boosting, sparse-input handling.
- Sentence-BERT (Reimers & Gurevych, 2019): semantic similarity beyond keywords.
- Key challenges: label noise, data sparsity, vocabulary drift, fairness.

3. DATA & PREPROCESSING

- Col. normalisation — snake_case, strip BOM artefacts
- Null-token replacement — 22 placeholder strings → NaN
- Drop 8 cols with ≥92% nulls (locations, age_req., ...)
- Binary presence flags for 5 optional resume sections
- Parse 12 list-type cols from stringified Python lists
- Date parsing → total_years_experience (capped 40 yr)
- Outlier clipping: matched_score ∈ [0,1], exp. ∈ [0,40]
- Duplicate removal; median/mode imputation

Metric	Value
Raw rows	9,544
Raw cols	35 → 30
Target	matched_score [0,1]
Eng. features	16

4. FEATURE ENGINEERING

Family	Features
Jaccard	skill_jaccard, related_skill_jaccard, position_jaccard
TF-IDF Cosine	skill_vs_resp, relskill_vs_resp, pos_vs_title, obj_vs_resp
Counts	num_skills, num_positions, num_degrees, num_related
Flags	has_address, has_objective, has_languages, has_cert, has_activity
Temporal	total_years_experience, has_valid_experience
Embeddings	sem_full, sem_skills, sem_position, sem_objective

5. MACHINE LEARNING MODELS

Three ensemble regressors trained on an 80/20 split (random_state=42) to predict the continuous matched_score target.

Model	n_est	depth	lr
XGBoost	150	4	0.05
Random Forest	150	6	—
Gradient Boosting	100	3	0.05
• Features: all engineered numerics; NaN filled with 0.			
• Best model (highest R²) saved for Predict mode.			
• K-Means (K=2–10): StandardScaler → fit → PCA 2D viz.			

6. RESULTS & DISCUSSION

0.847 R² XGBoost	0.078 RMSE	0.934 AUC-ROC
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Model	MAE↓	RMSE↓	R²↑
XGBoost ✓	0.052	0.078	0.847
Random Forest	0.059	0.084	0.821
Grad. Boosting	0.061	0.087	0.808
Baseline (mean)	0.142	0.186	0.000

Binary Classification (threshold = 0.50):

Accuracy	F1 Score	AUC-ROC
88.4%	0.861	0.934

Top Features (XGBoost gain):

skill_jaccard	0.94
sem_skills_sim	0.88
relskill_vs_resp_cos	0.81
skill_vs_resp_cos	0.75
sem_full_sim	0.68
position_title_jaccard	0.55
total_years_exp	0.42

K-MEANS CLUSTERS (K=4)

Cluster	Profile
0	High-skill specialists — top match scores
1	Generalists — broad experience, moderate match
2	Entry-level — few skills, many missing fields
3	Certified — active certs, higher edu. recency

7. DASHBOARD ARCHITECTURE

Tab / Mode	Capability
Tab 1 — Overview	KPIs, score dist., job/skill charts, correlation heatmap, tab
Upload	CSV or Excel; immediate schema preview
Discover	dtype profile, missing-value bar chart, sample rows
Preprocess	null threshold, fill strategy, dedup; stores clean df
Train	target selector, model checklist, test-size & threshold slide
Predict	batch score; auto-train if no model; download CSV
Cluster	elbow curve, PCA scatter, profile table, size chart

8. CONCLUSIONS

- XGBoost (R²=0.847, AUC=0.934) is the best model.
- skill Jaccard and semantic similarity are top predictors.
- Hybrid features (lexical + semantic) outperform either alone, confirming vocabulary-mismatch as the core NLP challenge.
- K-Means reveals 4 actionable candidate segments for talent-pool analysis.
- No-code dashboard enables full pipeline execution on any CSV without writing code.

9. FUTURE WORK

- Fine-tune BERT on labelled resume–job pairs for deeper semantic matching.
- Replace algorithmic labels with recruiter-assessed outcomes (hired/not).
- Add SHAP explainability panel to dashboard for per-candidate transparency.
- Fairness auditing: demographic-parity checks on protected attributes.
- Production hardening: model registry, multi-user auth, concept-drift alerts.

10. REFERENCES

Chen & Guestrin (2016). XGBoost. KDD '16.
Fernández-Delgado et al. (2014). Do we need hundreds of classifiers? JMLR.
Pedregosa et al. (2011). Scikit-learn. JMLR 12.
Qin et al. (2018). Person-job fit. SIGIR '18.
Reimers & Gurevych (2019). Sentence-BERT. EMNLP '19.
Robertson & Zaragoza (2009). BM25. Found. & Trends IR.
Siting et al. (2012). Job recommender systems. ICCSE.

11. TOOLS & METRICS REFERENCE

Tool	Role
Pandas / NumPy 2.x	Data wrangling
Scikit-learn 1.5+	Models, metrics, scaling
XGBoost 2.1+	Primary ML model
Sentence-Transformers	Semantic embeddings
Plotly Dash 2.x	Interactive dashboard
ReportLab 4.x	PDF generation

Metric	Formula	Best
MAE	mean y−■	↓ 0
RMSE	√mean(y−■)²	↓ 0
R²	1−SSres/SStot	↑ 1
F1	2·P·R/(P+R)	↑ 1
AUC	Area under ROC	↑ 1